

Advances in Science, Technology & Innovation
IEREK Interdisciplinary Series for Sustainable Development

Pir Mohammad · Shahfahad · Jesús Rodrigo Comino ·
Abraham R. Matamanda *Editors*

Geographic Approaches to Climate Change and Mitigation: Urban and Rural Perspectives (Volume 1)

Advances in Science, Technology & Innovation

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ASTI series has now been accepted for Scopus (September 2020). All content published in this series will start appearing on the Scopus site in early 2021.

Pir Mohammad · Shahfahad ·
Jesús Rodrigo Comino · Abraham R. Matamanda
Editors

Geographic Approaches to Climate Change and Mitigation: Urban and Rural Perspectives (Volume 1)

A Culmination of Selected Research Papers from the
International Conference on Geographic Perspectives on Climate
Change Mitigation in Urban and Rural Environment—1st
Edition—University of Salento, Italy

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ISSN 2522-8714 ISSN 2522-8722 (electronic)
Advances in Science, Technology & Innovation
IEREK Interdisciplinary Series for Sustainable Development
ISBN 978-3-031-92118-6 ISBN 978-3-031-92119-3 (eBook)
<https://doi.org/10.1007/978-3-031-92119-3>

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The Editors warmly thank all the Reviewers who have contributed their authority to the double-blind review process, to ensure the quality of this publication.

Preface

As we travel through the intertwined narratives of climate change mitigation in urban and rural environments, it is essential to recognize the urgency and complexity of the challenge we face. Climate change is not merely an environmental issue; it is a multifaceted crisis that touches every aspect of human life, economy, and society. From the bustling streets of metropolitan areas to the serene landscapes of rural communities, its effects are felt universally, albeit in diverse ways.

This book, *Geographic Approaches to Climate Change and Mitigation: Urban and Rural Perspectives (Volume 1)*, is an outcome of the selected papers presented at the 1st International Conference on Geographic Perspectives on Climate Change Mitigation in Urban and Rural Environments (GCUE 2024) held from 25 to 27, June 2024, at the University of Salento, Italy. This conference is a shared commitment to exploring these varied impacts and to highlight innovative strategies that can be employed across different settings. Our aim is to close the gap between urban and rural approaches, encouraging an understanding of how climate solutions can be tailored to the unique characteristics of each environment. This edited book is divided into two volumes. This volume will be divided into six parts, consisting of 33 chapters. Each chapter is devoted to distinct yet interconnected aspects of the subject matter.

This book is not merely an assortment of academic papers; it represents a collective endeavor to understand and address the complexities of urban and rural environments. Each contributor brings a wealth of knowledge and insight, sharing case studies and research that illuminate successful initiatives, and the lessons learned along the way. This collaborative effort is a testament to the power of interdisciplinary work and the importance of engaging multiple perspectives in addressing global challenges.

As you go through the chapters that follow, I encourage you to reflect on your own role in climate conversation. Whether you are a researcher, a student, a policymaker, or a community advocate, each of us has a part to play in shaping the future of our planet. The insights and strategies presented here are intended not only to inform but also to inspire action.

In closing, I would like to express my gratitude to all the contributors, reviewers, and supporters who have made this book possible. Together, we can forge a path toward a more sustainable and resilient future, one that honors the unique contributions of both urban and rural environments in the fight against climate change.

Hung Hom, Hong Kong

Pir Mohammad

Acknowledgments

The completion of this book, *Geographic Approaches to Climate Change and Mitigation: Urban and Rural Perspectives (Volume 1)*, represents the collective effort of many individuals and organizations whose contributions have been invaluable. First and foremost, we would like to express our deepest gratitude to the contributing authors, whose expertise and insights have enriched this volume. Your dedication to advancing the understanding of climate change mitigation in diverse contexts is evident in every chapter. We appreciate the time, effort, and passion you invested in your research and writing.

We extend our heartfelt thanks to the reviewers, whose constructive feedback and critical evaluations have greatly enhanced the quality of this book. Your attention to detail and commitment to excellence have ensured that the content is both rigorous and accessible.

We acknowledge and appreciate the efforts of IEREK in organizing the conference in collaboration with University of Salento, Italy and giving a platform of various researches worldwide to come together and discuss the issues of urban and rural environments in regard to climate change and its mitigation. We also appreciate and acknowledge the hosts and co-hosts of the conference for organizing it. We are extremely grateful to all our supporting organizations of the conference.

A special thank you goes to our editorial team and publishing staff, whose professionalism and guidance have been instrumental in bringing this book to fruition. Your expertise in navigating the publishing process has been greatly appreciated.

To our families and friends, thank you for your understanding and encouragement during the long hours spent working on this project. Your support has been a source of strength and motivation.

Finally, we would like to acknowledge the communities, both urban and rural, who are at the forefront of climate change impacts. Your resilience, creativity, and commitment to sustainability inspire us all. This book is dedicated to you and to those who strive to create a more equitable and sustainable future for our planet.

Introduction

Climate change is one of the most pressing challenges of our time, affecting ecosystems, economies, and communities across the globe. From rising sea levels and increased frequency of extreme weather events to shifts in agricultural productivity and public health crises, the ramifications of climate change are profound and far-reaching. As the urgency to address these issues escalates, the need for effective and innovative climate change mitigation strategies becomes necessary.

This book, *Geographic Approaches to Climate Change and Mitigation: Urban and Rural Perspectives (Volume 1)*, aims to explore the diverse approaches to climate change mitigation through the lens of geography, emphasizing the distinct contexts of urban and rural areas.

Urban environments are at the forefront of the climate crisis. They are characterized by high population density and significant infrastructure, facing unique challenges in mitigating climate change. The challenges faced by urban centers are multifaceted: from managing waste and transportation emissions to ensuring equitable access to green spaces and sustainable resources. Cities are often hotspots for greenhouse gas emissions, but they also serve as centers for innovation, policymaking, and community mobilization. They possess the infrastructure, economic resources, and human capital necessary to implement transformative change. By harnessing technology, fostering public engagement, and promoting sustainable practices, urban areas can lead the way in mitigating climate change impacts.

In addition, rural areas are often overlooked in climate discussions even though they play a critical role in both climate change mitigation and adaptation. Though they contribute less to global emissions, rural regions are uniquely vulnerable to the effects of climate change, including altered precipitation patterns, extreme heat, and threats to food security. These communities are often dependent on natural resources for their livelihoods, making them particularly sensitive to environmental changes. Understanding the challenges faced by rural areas is essential for creating strategies that not only reduce emissions but also enhance resilience and support sustainable development.

This book seeks to close the gap between urban and rural perspectives, recognizing that both environments must work together to achieve meaningful climate action. The interplay between urban and rural areas is complex; for instance, urban demand for resources can drive unsustainable practices in rural regions, while rural ecological health can directly impact urban quality of life. By encouraging collaboration between urban and rural stakeholders, we can develop integrated strategies that use the strengths of both environments.

Throughout this volume, readers will find a diverse range of perspectives from scholars, practitioners, and policymakers who are actively engaged in climate change mitigation. This volume explores six broad topic areas: Applications of Climatology in Urbanism, The Role of Geography in Climate Change Mitigation, Meteorology and Disaster Prevention in Urban and Rural Environments, Urban Planning and Climate Change Mitigation, Rural Land Use and Climate Change Mitigation, and Geographic Mapping and Innovative Systems for Rural Fragmentation Protection. Each chapter addresses key questions that are vital to our collective understanding: How can urban areas leverage their resources and infrastructure to spearhead climate mitigation efforts? What innovative practices are rural communities employing to enhance resilience while

minimizing their carbon footprints? How can policies be designed to promote synergy between urban and rural stakeholders, ensuring a cohesive approach to climate action?

The contributions in this book are collected in the form of case studies and empirical research, showcasing successful initiatives and lessons learned from various geographic contexts. From urban green infrastructure projects aimed at reducing heat island effects to rural agricultural practices that enhance soil health and carbon sequestration, the chapters illustrate the breadth of strategies available. Additionally, theoretical frameworks are presented to help contextualize these practices within broader climate change narratives.

Ultimately, this book aims to work as a resource for researchers, students, and decision-makers committed to combating climate change. By integrating geographic perspectives and emphasizing the unique challenges and opportunities within urban and rural environments, we hope to inspire new ideas and collaborative efforts. Our goal is to promote resilient communities that are not only equipped to face the challenges of a changing climate but also actively contribute to a sustainable and equitable future for all.

Pir Mohammad

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Climate Maps and Urban Planning in Search of a Reliable Framework for Adapting Cities and Communities to the Impacts of Climate Change

Rosalba D'Onofrio, Giorgio Caprari, Roberta Cocci Grifoni, Francesco Conti, Graziano E. Marchesani, Simone Malavolta, Ilenia Pierantoni, and Massimo Sargolini

Abstract

The urban environment is particularly vulnerable to the catastrophic effects of extreme weather events, making it crucial to have a reliable framework in place. Numerous studies have shown how urban geometry, building materials, and vegetation can affect indoor and outdoor comfort. Integrating climate assessments into urban planning can be challenging. This involves linking climate issues identified in an urban context with broader urban planning considerations and tools. The PNRR-Ecosystem of Innovation—Vitality Project aims to explore the connections between urban climatology, urban form, and social relations. The project seeks to use climate maps to inform urban planning and design. This contribution documents

the development of knowledge that has led to the investigation of climate maps using two documented methodologies: UCMaP (Urban Climate Map) and LCZ (Local Climate Zones). The goal is to identify strengths and weaknesses to create a potential path for innovation in classifying elementary climate units (from LCZ to CAU.us—Climate Adaptive Urban Units) and develop guidelines and design proposals at the urban and detailed climate-proof scale.

Keywords

Local climate zones · UCMaP · Urban planning · Climate adaptation · Urban Design

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1 Introduction

Urban planning has a significant impact on climate change-related actions (Abougendia et al. 2020; Carter et al. 2015; Hurlimann et al. 2021). However, the application of urban climatology in urban planning is still limited (Aslam and Rana 2022). The existence of this gap in the literature can be attributed to (1) the different language and knowledge between urban climatologists and urban planning academics and professionals (Eliasson 2000), (2) the difficulty in finding morphological and climatic data at the urban scale, and (3) the limited application tools for translating climate knowledge to the operational scale (Eliasson 2000; Yin et al. 2022). These obstacles are compounded by (4) the specificity of research findings and (5) the paucity of experience attempting to standardize processes to quantitatively describe urban climate (Chien et al. 2016).

Currently, scientific research includes two conceptual frameworks for assessing urban climate that may be relevant to urban planning: the Urban Climatic Map (UCMaP) system (Ren et al. 2011) and local climate zones (LCZs). (Stewart and Oke 2009, 2012).

The UCMaP (Urban Climate Map) is a cartographic representation of climate information and the translation of climate assessment results into a planning language. The methodology behind it is the work of the Association of German Engineers (VDI), which in the 1990s published a national standard, VDI 3787-Part 1 (VDI 1997), to guide the application of the UCMaP in local practice. The UCMaP system usually consists of two components: the Urban Climate Analysis Map (UC-AnMaP) and the Urban Planning Recommendations Map (UC-ReMaP). The UC-AnMaP collects meteorological, planning, land use, topographic and vegetation information. Their interrelationships and impacts are spatially analysed and evaluated (Ren et al. 2013) through the identification of climatic areas called “climatopes”. The UC-ReMaP includes climate planning recommendations for wind, vegetation, water and urban morphology (Ren et al. 2013). The UCMaP uses mesoscale and microscale weather simulation models or empirical data from field measurements (Chen et al. 2017; Zhongli and Hanqiu 2016). There are two main problems with the generation of UCMaPs. First, simulation efficiency is limited by computational accuracy and complicated operational techniques. Second, measured empirical data usually cannot fully represent the different surface conditions associated with multiple urban morphologies (Yin et al. 2022). To address these limitations, it seems essential to rely on a better technical system using a land surface classification scheme suitable for urban planners and a more standardized urban climate assessment model (Jin et al. 2020).

The Local Climate Zones (LCZs) proposed by Stewart and Oke (2012) are aligned with this objective. They represent a relatively novel conceptual framework for urban climatology, as they have succeeded in representing both the characteristics of climate and urban morphology at the city level worldwide. The LCZ system is based on the application of generalized knowledge of building forms and land cover and comprises 17 distinct classes. The thermal climate of each class is determined by its surface characteristics, including thermal, radiative, geometric, metabolic and surface cover properties.

LCZs 1-10 are designated for “built” typologies, which may be defined as the expression of urban form (Eldesoky et al. 2022). LCZs A-G are designated for “natural” typologies, which represent permeable land cover, water bodies, and vegetation. The standardization of LCZ definitions and classifications allows for comparisons to be made within and between cities. The LCZ system operates on a scale between cities and neighbourhoods and strongly correlates with land use and building functionality. This allows for the representation of different local climates (Stewart and Oke 2012). The advent of big data and GIS spatial processing techniques has led to a continuous evolution in research methods, resulting in a multitude of experiments conducted at varying scales

(Quan and Bansal 2021; Zheng et al. 2018). To date, almost 300 cities worldwide have conducted urban climate studies in the context of LCZs (Zhou et al. 2022). Notable projects such as the World Urban Database and Access Portal Tools - WUDAPT (Demuzere et al. 2021; Ren et al. 2016), MapUCE (Masson et al. 2015), and URBAN-FLUXES (<http://urbanfluxes.eu/>) have consistently demonstrated advancements in data optimization. Moreover, the development of sophisticated techniques, including deep learning algorithms, pattern simulation, and neural networks (Lau et al. 2023), has led to the integration of these methods in LCZ mapping.

Nevertheless, there are still some shortcomings that require attention. Despite the LCZ's efficacy in educating urban planners and designers on the impact of urban form and function on urban temperatures and heat wave risks, there remains a paucity of substantial experiences to inform urban planning (Lelovics et al. 2014; Ren et al. 2011). One of the most critical aspects is the size of the LCZ methodology's local units, which serve as the minimum units of investigation and evaluation (Perera and Emmanuel 2018). Traditional urban planning is conducted at multiple levels, from the neighbourhood to the urban and regional scales. At present, the most prevalent approach to LCZs entails the categorization of urban blocks as homogeneous units (Quan and Bansal 2021). Accordingly, to investigate the impact on extensive urban areas, it may be necessary to identify a compromise between block-level urban areas and larger planning zones (Perera and Emmanuel 2018). In comparison to the Urban Climatic Map system, LCZs offer a number of notable advantages. These include the requirement for less data, the presence of universal classification criteria, the capacity for comparisons between cities, and enhanced accessibility for non-professionals (Chen et al. 2019; Zhongli and Hanqiu 2016). To date, no studies have been conducted to compare and contrast the two methodologies, to ascertain their respective differences, to explore their potential for combination and to identify any potential mutual benefits (Yin et al. 2022).

This article proposes an analysis of the literature on the experiments related to climate maps, with the goal of:

- developing a reflection on the performance of climate maps created with UCMaP and LCZ methodologies, with a view to verifying their usefulness for urban planning and the opportunity for their hybridization;
- proposing a methodology for the construction of a climate map that will be tested in the PNRR Vitality project, Spoke 6 “Innovation and safeness in living environments in the digital and green transition”, Work Package 1: “CONTEXT: INDOOR AND OUTDOOR AT THE CENTER OF THE DIGITAL AND GREEN TRANSITION OF LIVING ENVIRONMENTS”. This will support the Project in providing urban planning and urban design

with a useful and complete tool for tackling climate adaptation issues by combining the LCZ methodology with some of the UCMaP's capabilities.

2 Materials and Methodology

A systematic review was conducted using the Scopus database with the following keywords: "UC-AnMap/UC-Map and urban planning", "UC-ReMap and urban planning", "Climatopes and urban planning" and "LCZ Map and urban planning". This resulted in the selection of 188 scientific articles. The initial search, conducted with the keywords "UC-AnMap; UC-ReMap and urban planning", yielded a total of 15 articles. Subsequently, an additional 13 articles were identified through the extension of the search to include the keywords "climatopes and urban planning". A search for "LCZmap and urban planning" yielded 160 articles.

Following a review of the abstracts and the exclusion of articles not written in English, a total of 56 articles were selected based on the content of the introduction. A comprehensive reading of the full papers enabled the selection of an even smaller number of significant articles, totalling 21. From these, four were selected for the UCMaP methodology and only three for the LCZ methodology. The restricted selection process involved the exclusion of articles that solely focused on the expression of interest in the relationship between urban planning instruments and climate surveys, without establishing a clear and effective connection between the two.

The UCMaP experiments that we have gathered include those conducted in the cities of Arnhem (Ren et al. 2012), Lisbon (Alcoforado et al. 2009), Kaohsiung (Ren et al. 2012) and Beijing (He et al. 2015). In the case of LCZ methodology experiments, the experiences that we considered most significant were those of some Chilean cities involved in the Research Projects "Conicyt/Fondecyt 1,130,305 (2013–2016)" and "ANID/Fondecyt initiation 11,180,990", the city of Bogotá (Jiménez-Ariza et al. 2023) and the city of Colombo in Sri Lanka (Perera and Emmanuel 2018). To develop this process, we used the following keys:

- objectives of climate maps;
- climatic and urban planning analyses fielded;
- classification and sizing of climate areas;
- relationship with urban planning.

At the conclusion of the study, the findings were employed to set a potential methodology for the delineation of the contents of a morpho-climatic map. It was proposed that this map could exert influence and effect upon urban planning and urban design within the context of the Vitality Project.

Indeed, one of the objectives of this PNRR (National Plan for Recovery and Resilience) initiative is to delineate pivotal design parameters that emphasize the interrelation between the building scale and the "climate-adaptive urban unit scale" (CAU.u). This will be achieved through the application of quantifiable factors to facilitate the development of mitigation and adaptation strategies that enhance the conditions of comfort, safety, energy efficiency and environmental sustainability of indoor and outdoor spaces within the settlement systems of Central Italy.

3 Results

In examining the cities that have employed the UCMaP methodology, it became evident that a common objective exists, namely, to gain insight into the existing urban climatic and environmental conditions through the utilization of a range of climatic parameters. This information is subsequently employed and interpreted from an urban planning perspective, with the objective of creating the city's climate map and generating urban planning recommendations. In terms of climate objectives, the aim is to reduce the heat-island effect and improve urban ventilation. For instance, the cities of Lisbon, Beijing and Arnhem prioritize these objectives, whereas Kaohsiung is focused on reducing the effects of the heat island and rising temperatures in general. A comprehensive cognitive framework is initially established in order to construct the UC-An Map. This includes the collection of meteorological data (such as average temperatures, precipitation, wind speed and solar radiation), as well as data on land use, topography and vegetated areas. Subsequently, the second phase combines the aforementioned information levels to understand their effects on the energy and heat balance at the meso- and/or micro-climatic level, depending on the scale of interest. Additionally, other information concern urban morphology parameters, such as building density, which affects the roughness of the urban soil, which in turn conditions ventilation and the built environment's capacity to absorb heat; the percentage of land cover; proximity to open spaces, etc. In the case of Arnhem, the analyses covered a number of key areas, including land use, the construction of the digital terrain model, basic meteorological data with particular reference to wind conditions, and the role of urban morphology in relation to thermal aspects. In the case of Lisbon, the cognitive analyses were oriented towards the construction of a map of the different built densities (high, medium, low, and very low) and a ventilation map. Subsequently, a co-occurrence matrix was constructed between the various land cover/urban density/ventilation classes with the objective of identifying climate-response units (HCR). In the case of Beijing, the cognitive apparatus is based on

the recognition of the “thermal load map,” which expresses the intensity of heat stored or emitted by urban areas. This parameter is contingent upon the topographic height, the sky factor, the presence of green space and water. The presence of these components is then entered into a geographic information system (GIS) and subjected to assessment. A comparable methodology is employed in the construction of the ventilation potential map, whereby a multitude of fundamental layers are integrated, encompassing building height, roughness length, building coverage, and woodland terrain. A comparable climate assessment was conducted in Kaohsiung. In this instance, the cognitive analyses encompassed the collation of meteorological data, the examination of land use and planning, the analysis of population data, the assessment of UHI intensity, the investigation of natural landscapes, and the evaluation of water bodies.

A climate assessment matrix was developed based on the impacts of these topics on thermal load (TL) and dynamic potential (DP). This matrix reports the effects on thermal load, dynamic potential, wind, etc. In the case of Arnhem, the areas covered by the climate assessments included: transportation areas (railway, road, airport); built-up areas (residential, commercial and catering, public space, socio-cultural facilities, building site); semi-built-up areas (landfill, cemetery, building site and others); leisure (park, sports ground, holiday resort); agriculture; forest and natural areas; water bodies (Rhine-Meuse river, recreational water areas, alluvial areas, wet field water channel). In the case of Lisbon, the areas subject to climate assessment are defined by their physical morphologies, including hilltops, the northern plateau, the southern plateau, the main valley beds, the Tagus bank, green areas, and so forth. Regarding Beijing, the urban areas were classified into seven categories, with the prevailing environmental and settlement characteristics corresponding to the climatic features. In the case of Kaohsiung, three areas of the city were identified, corresponding to districts for which the principal environmental characteristics and climatic risks are summarized.

With regard to the construction of the UC-AnMap, the city of Arnhem has identified four distinct climate zones, which correspond to built-up areas, green spaces, water, and transportation infrastructure. The potential for thermal load (TL) and dynamic potential (DP) have been identified for these areas. In the case of Lisbon, the identification of HCR units is based on delineating seven distinct areas, which are characterized by specific topographical features, population density, vegetation cover, and climatic characteristics. In the case of Beijing, a platform of climatic information for planning was created with the objective of assisting decision-making. The five “Planning Zones” are distinguished as follows: Zones of High Climate Value (with respect to fresh air ventilation paths); Zones of Climate

Value (green wedge effects); Moderately Climate-Sensitive Zones (densely built-up areas with a high heat load and low ventilation); and Highly Climate-Sensitive Zones (densely built-up areas with a very high heat load and extremely low ventilation). In conclusion, the recommendations for urban planning in the case of Arnhem present a summary of the main climatic conditions and the major problems and actions to be promoted for each area type. To illustrate, the area type designated as “High thermal load and very low dynamic potential” is indicative of commercial and residential zones. In this instance, the primary issue identified is the urban heat island effect. To address this, the proposed measures include the creation of an air path to facilitate the passage of cool breezes through the city centre, as well as the control of further urban expansion.

In the case of Lisbon, the recommendations for urban planning with reference to the different HCR Units are particularly detailed. To illustrate, the HCR Unit “High-Density Urban Areas of the Southern Plateau,” which encompasses the concentrated urban fabric of the historic centre and a significant portion of the city constructed before the 1950s, is guided by the following primary urban planning principles: (1) it is recommended that an increase in built density in the valley bottoms be avoided without a detailed study of the consequences on the ventilation of the valley; (2) it is recommended that the H/W ratio (i.e., the ratio between the height (H) of buildings and the width (W) of streets) be kept below 1 in cases where urban development is currently taking place; (3) it is recommended that vegetated areas, including roof gardens, be maximized; (4) it is recommended that light colours and materials with low thermal admittance be selected in the case of building renovation. A similar level of detail can be observed in the case of Beijing, with reference to the different planning areas. In Kaohsiung, three levels of planning actions were developed: city, urban, and neighbourhood scales. For each urban district, potential control measures were devised and summarized according to criteria such as greenery, shading, albedo, anthropogenic heat release, air exchange, and air pollution. The recommended course of action comprises three distinct categories: improvement actions, mitigation actions, and protection actions.

In the case of the LCZ methodology, the cases examined concern the case study of 21 Chilean cities involved in the Research Project “Conicyt/Fondecyt 1,130,305 (2013–2016) and ANID/Fondecyt initiation 11,180,990 project”, as well as the city of Bogotá and the city of Colombo (Sri Lanka). In the case of Chilean cities, the primary objective was to mitigate the intensification of extreme events, including daytime heat associated with increased heat waves, nighttime heat, and tropical nights. Local climate zones (LCZs), delineated by urban design variables such as building density and

height, were correlated with other urban attributes influencing climate, including surface temperature and vegetation. The recognition of LCZs was conducted through the photointerpretation of images accessible in Google Earth Pro software (2021), in accordance with the recognition criteria and utilization of colours proposed by Wudapt. Subsequently, an investigation was conducted into the correlation between surface temperature and vegetation within the designated LCZs, utilizing the Normalized Vegetation Difference Index (NDVI). LCZ3 is the most prevalent local climate zone in urban areas worldwide. It is typified by residential urban areas comprising detached houses with one to three storeys, constituting between 20 and 45% of the urban area in different cities. This LCZ was subsequently selected to investigate the correlation between temperature and vegetation activity in six cities situated at varying latitudes. This is due to the fact that it is one of the typologies that is least affected by the presence of shadows cast by buildings. To enhance the climatic conditions of urban areas, a series of urban planning measures are currently being developed. These measures aim to modify the climatic behaviour of cities by modifying the LCZ categories to which they belong. In this regard, a number of factors have been identified which urban planning tools should address. These include the height of buildings, which is associated with the proportion of visible sky and, therefore, with the amount of sunlight and shadows that affect a given area over a year; the building coefficient, which influences the density of buildings; the distance between adjacent buildings, which determines the amount of available space and is therefore associated with sunlight and shade; and the role that vegetation plays in reducing temperatures. Instead of this, the recommendations for urban design at the detailed scale are organized in a table based on three principal parameters: colours, materials and density. The potential impacts of these three parameters are identified, and recommendations are made to avoid the most harmful effects.

In the case of Bogotá, the necessity to address the issue of heavy rainfall and urban water drainage has prompted the utilization of LCZs and LCZSs (cell subdivisions of 20×20 m). This approach enables the assessment of the requirements and potential for the provision of ecosystem services across the city's diverse locations, with the aim of identifying the most suitable NBSs for water runoff management. The territory is classified into six LCZs and 18 LCSZs based on eight variables with a significant relationship with temperature. These variables include altitude, residential use, tree canopy, NDVI, building density, water table, distance to main roads and distance to transport stations. For each subdivision of the LCZs (which were divided into three cells based on the amount of annual rainfall), recommendations are defined in accordance

with the potential benefits and feasibility of implementation. Firstly, priority and opportunity scores are defined for each NBS type in each cell, considering the NBS type's potential to contribute to ecosystem services. The analysis provides a qualitative score on a scale of zero to three, based on an assessment of the feasibility of the results. Subsequently, the NBS are filtered by identifying potential impediments to their implementation in the same cell and selecting the types with a higher score if area requirements or similar characteristics and functions might limit simultaneous use. This process yields a list of NBS for each cell, which assists in identifying stormwater management processes (i.e., detention, retention, transport and infiltration) by considering the characteristics and information provided in the literature (Woods-Ballard et al. 2016). In the case of Colombo, Sri Lanka, the approach used involved matching LCZs to urban blocks as the area of survey and climate assessment. Each block was mapped and merged with neighbouring blocks exhibiting similar values for sky view factor (SVF), aspect ratio (H/W), building surface fraction (BSF), impervious surface fraction (ISF), pervious surface fraction (PSF), height of roughness elements (HRE), terrain roughness class, surface admittance, albedo and anthropogenic heat flux. Subsequently, each isolate was assigned the corresponding LCZ and, in some cases, a subclassification was also required. A preliminary subclassification of 46 LCZ combinations, including primary LCZs, was conducted. Subsequently, to avoid excessive fragmentation, 31 subclasses were employed, comprising 16 primary and 15 subclasses. The simplified mapping of the land use-based urban system, with some consideration of urban activities, allows for a more straightforward interpretation of the climate. It is proposed that LCZs should be designated as a fundamental unit for urban planning. This is aimed at promoting sustainable urban development and preventing the occurrence of intensive urban development under urban planning instruments that encourage land use (Table 1).

4 Discussion

The principal objective of the intervention strategies and urban planning recommendations outlined in a climate map is to facilitate a transformation in urban settlement characteristics or the preservation of specific territorial conditions that can safeguard them from the catastrophic consequences of climate change. Both described methodologies aim to create climate maps, but they have different practical outcomes for urban planning. UCMaps provides climate information at the regional or urban scale for planners and policy-makers. This is achieved by grouping the data according to the scale of different climate zones, which are called

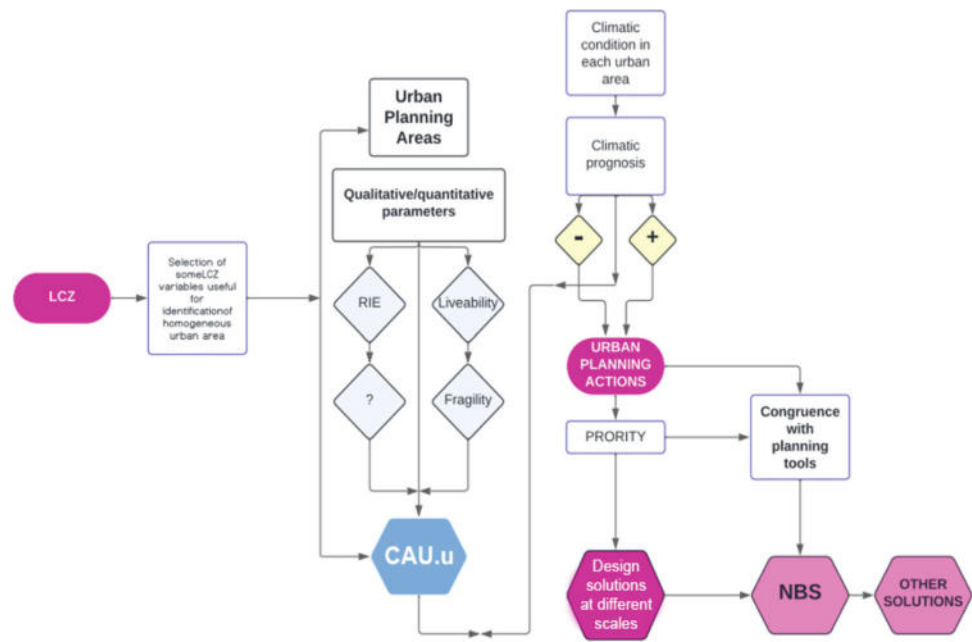
Table 1 Summary comparison between UCMaP and LCZ methodologies

	Urban climate map (UCMaP)	Local climate zones (LCZ)
Definition	A cartographic representation of climate information, integrating climate assessments into urban planning	A classification system representing urban morphology and climate characteristics using 17 standardized classes
Components	Consists of Urban Climate Analysis Map (UC-AnMaP) and Urban Planning Recommendations Map (UC-ReMaP)	Composed of 17 classes, including built and natural typologies, based on building forms and land cover
Scale	Operates on regional or urban scales, focusing on broader climate zones called “climatopes”	Functions at city and neighbourhood levels, allowing detailed local climate representation
Data requirements	Requires detailed meteorological, planning, land use, topographical, and vegetation data	Uses generalized data on building forms and land cover
Methodology	Utilizes mesoscale and microscale weather simulation models or empirical data from field measurements	Employs standardized classification criteria, often using satellite data and GIS techniques
Current use and potential for innovation	• Used for creating climate maps and generating urban planning recommendations to reduce heat islands and improve ventilation • Provides comprehensive climate information for planners and policymakers • Established tradition in urban planning	• Useful for predicting urban fabric transformations and assessing climate adaptation needs • Standardized morphological classification allows comparisons across cities • Easier for non-professionals to understand • Could be enhanced by integrating more parameters for a holistic urban planning approach
Limitations	• Simulation efficiency is limited by calculation accuracy and operational complexity • Empirical data may not fully represent diverse urban morphologies • Primarily serves as an informational base with limited action prioritization	• Limited significant experiences in guiding urban planning • Determination of local unit dimension for traditional urban planning scales • May require a compromise between block-level and larger planning zones for broader impact
Studies and applications	Arnhem (Ren et al. 2012), Lisbon, Kaohsiung (Ren et al. 2012) and Beijing	Chilean cities involved in the Research Projects “Conicyt/Fondecyt 1,130,305 (2013–2016)” and “ANID/Fondecyt initiation 11,180,990”, Bogotá (Jiménez-Ariza et al. 2023), Colombo (Perera and Emmanuel 2018)

climatopes. This approach allows for a reduction in thermal load and an increase in the dynamic potential of a particular climate zone. UCMaPs have a well-established tradition in urban planning, yet their use provides only an informational basis. The generated maps are regarded as a significant visual instrument for planners to convey information to political leaders and the general public (Ren et al. 2011). The recommendations provided are largely confined to a single set of potential courses of action. A further limitation is that they rarely contemplate a classification of actions based on priority in terms of effectiveness and feasibility relative to the urban context. Conversely, LCZs have yet to be widely employed in urban planning, and their potential for enhancing urban climate resilience is significant. If they could be used to describe the climate performance of homogeneous territorial areas, they could provide additional knowledge and assessment for urban planning, particularly in the context of zoning. The utilization of LCZs offers a distinctive advantage, as exemplified by the city of

Colombo. This lies in the capacity to anticipate potential transformations of urban fabrics by elucidating the inherent risks and opportunities associated with diverse urban policies, both extant and prospective (Perera and Emmanuel 2018). Furthermore, as evidenced by the case of Chilean cities, a change could be planned in the more problematic LCZ categories, which would have significant implications for urban planning. In the case of Bogotá, the designation of “planning” units can facilitate the decision-making process for implementing climate adaptation and mitigation actions utilizing NBS (nature-based solutions). These are selected based on the priorities identified in the various LCZs. Furthermore, they can assist in identifying strategic intervention areas and formulating potentially replicable environmental improvement interventions in other contexts (Jiménez-Ariza et al. 2023). The experiences of the LCZs discussed in this article illustrate the importance of selecting appropriate boundaries for the division of areas (Perera and Emmanuel 2018). It can, therefore, be concluded that

Fig. 1 Proposed methodological flowchart for integration of LCZs as urban planning. (By author)



the LCZ methodology is not particularly effective when attempting to assess the impact of climate change on extensive urban areas. A compromise between the urban and territorial scales and the block scale is likely to be necessary, as evidenced by the cases of Bogotá and Colombo. Furthermore, it would be beneficial for the LCZ methodology to be more closely aligned with the specific political and social contexts at the local level. It is recommended that studies on urban temperature focus on human activities rather than solely on climate classification based on physical properties (Yu et al. 2021).

In light of the findings of this comparison, which is by no means exhaustive, we have defined and proposed a potential further contribution to the debate regarding urban planning and design in the context of the Research Project Vitality. The proposal is to create a morpho-climatic map that integrates the standardization system of the LCZ (Local Climate Zone) map with various social, economic, environmental, and energy factors. This integrated map will help in gaining a comprehensive understanding of urban environments as complex ecosystems. It will enable the development of customized urban planning solutions to address the specific challenges and requirements of different urban contexts as shown in Fig. 1 representing the integration of cAU.u in the urban planning and design process. The LCZs will be clustered and

reconfigured as CAU.us (Climate Adaptive Urban Units), and a combination of intricate parameters and indicators will be employed to assess the urban environment, thereby enabling the formulation of bespoke urban planning solutions that can address the specific challenges and meet the distinctive needs of different urban areas. The parameters mentioned above encompass the quality of building structures; the proportion of residential, commercial, and industrial areas, green spaces, urban mobility and road infrastructure parameters, environmental parameters, and energy and building indicators. Conventional data-gathering methods will be integrated with sophisticated technologies such as Geographic Information Systems (GIS), parameterised processes and remote sensing to identify and quantify these parameters within the CAU.us framework. The CAU.us will be grouped, as shown in Fig. 2, according to their analogous behavioural patterns with respect to morpho-climatic characteristics, social and health vulnerability, primary urban functions, vegetation quality, energy efficiency, and other potential parameters. The resulting clusters will constitute discrete urban areas, and priority levels will be assigned on the basis of the identified vulnerabilities, feasibility, and compatibility with the local social and economic conditions. Subsequently, the requisite adaptation measures will be implemented.



Fig. 2 Clusterization process from LCZs to CAU.us in Ascoli Piceno (Italy) historical centre. (By author)

5 Conclusions

Climate maps have the potential to play a pivotal role in urban planning at various stages of the design process. From the experiences examined, UCMaps have been primarily used for recognition and information. However, it is possible that the LCZs, which urban planners currently underuse, could play a central role in designing adaptation actions if they incorporate a series of parameters and indicators that value other components of the urban system, such as functions, social and demographic components, in addition to physical ones. The PNRR Vitality Project can utilise LCZs categorized as CAU.us to direct urban regeneration or restructuring initiatives, thereby enhancing their efficacy. CAU.us can be employed to identify the primary critical issues and, consequently, select a series of beneficial actions to mitigate extreme climate events. It can also be used to ascertain the suitability of specific locations based on their climate, social, and environmental vulnerability. Furthermore, it can serve as a reliable instrument for evaluating the sustainability of land governance strategies. In conclusion, CAU.us has the potential to offer a more comprehensive and effective approach to urban planning. This could help in managing the environmental impact of urban areas, improving energy efficiency, and enhancing the quality of life and well-being of urban inhabitants.

Acknowledge Financial Support From PNRR MUR project ECS_00000041-VITALITY

Author Contributions Conceptualization, R.D. and R.C.G.; Formal analysis, R.D., G.C., R.C.G., F.C., G.E.M., S.M., I.P., M.S.; Methodology, R.D., R.C.G.; Supervision: R.D. and R.C.G.; Writing—review and editing, R.D., R.C.G. and I.P. All authors have read and agreed to the published version of the manuscript.

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